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ABSTRACT

The discharge kinetic characteristics of an ArF excimer laser are investigated by using a one-dimensional fluid model, coupled with an external circuit and laser gain modules. It is found that a higher F_2 concentration suppresses discharge breakdown, as indicated by a lower effective ionization coefficient and a higher breakdown field strength. Moreover, an increase in F_2 concentration elevates the densities of reactants F_2 and F^- but reduces the densities of Ar^* and Ar^+, which are all crucial for ArF formation. As a result of this competition, the ArF density is maximized at an intermediate F_2 concentration, leading to an optimal laser power. The maximized power conversion efficiency is predicted by the model to be 5.8% at an F_2 concentration of 0.15%.

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I. INTRODUCTION

Excimer lasers, which generate ultraviolet (UV) light through stimulated emission from rare gas–halogen mixtures, are widely utilized in applications such as semiconductor lithography,[1,2] material processing,[3,4] and refractive surgery.[5,6] Among them, the argon fluoride (ArF) excimer laser, emitting at 193 nm, is currently the primary light source for advanced lithography, particularly in technology nodes of 28 nm and above.[7] Commercial excimer lasers are generated by high pressure pulsed discharge pumping and require uniform discharge to ensure high laser quality. Optimizing the gas mixture ratio is crucial for improving discharge uniformity. These laser systems typically contain F_2 in concentrations ranging from 0.1% to 0.5%.[8] As the working gas for forming the upper laser level species ArF^*, an increase in F_2 concentration can, to some extent, enhance the production rate of ArF^*. Despite constituting only a small fraction of the gas mixture, even a modest increase in F_2 concentration can significantly impair discharge uniformity due to its highly electronegative nature and the halogen depletion instability.[9]

The experiments by Lou and Nagai *et al.*[10,11] have demonstrated that discharges at the optimal F_2 concentration are crucial for maximizing laser output power. Lou *et al.*[10] conducted a study on a discharge-pumped KrF excimer laser, demonstrating that the F_2 concentration significantly influences laser output power and pulse duration, with an optimal range of 0.20%–0.22% for maximum performance. This article also reported the reduction in F_2 concentration due to discharge-induced consumption. Nagai *et al.* conducted a study on a discharge-pumped ArF excimer laser, demonstrating that when the F_2 concentration exceeds the optimal level, the laser power sharply decreases due to photon absorption by F^- ions and the ArF quenching by F_2 molecules. Mathew *et al.*[12] demonstrated that the discharge stability time decreases markedly with increasing F_2 partial pressure, following an inverse square root relationship.

Compared to experimental studies, kinetic modeling offers a more convenient approach to control F_2 concentration parameters and effectively predicts the spatiotemporal evolution of quantities that are difficult to measure directly, such as electron density, ArF excimer density, and photon density. Previous models[13,14] of F_2-based excimer lasers often omit the direct dissociation reaction ($e + F_2 \rightarrow e + 2F$) in their chemical reaction tables. Kushner *et al.*[15] demonstrated that direct dissociation becomes a highly significant channel for F_2 consumption under high power density, even surpassing dissociative attachment ($e + F_2 \rightarrow F^- + F$) in rate. Nighan *et al.*[8] conducted a

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theoretical study on discharge-pumped KrF lasers, demonstrating that the F_2 direct dissociation reaction becomes a significant electron energy loss channel at F_2 concentrations above 0.3%. Petrov *et al.*[16] found through simulations that when the F_2 concentration exceeds 1%, the electron energy distribution function significantly deviates from the Maxwellian distribution. The increase in F_2 concentration raises the mean electron energy by depleting low-energy electrons (<10 eV) through F_2 attachment. Gu *et al.*[17] reported the effect of F_2 concentration on the Ar_2F^* emission intensity in a He–Ar–F_2 mixture.

These studies reveal the critical role of F_2 molecules in discharge dynamics, discharge uniformity, and laser power output. However, many previous simulation studies have adopted simplified modeling approaches, such as decoupling the excitation source or the laser module. The operation of electrically pumped excimer lasers involves multiple interconnected physical processes, including high voltage pulse generation, discharge breakdown, plasma chemical reactions, and radiative emission, which necessitate a comprehensive and multi-physically coupled modeling approach.

In this work, a one-dimensional (1D) fluid model integrating an external circuit module and a laser gain module is established for a pulsed discharge-pumped ArF excimer laser. The model is designed to operate under a gas mixture of Ne/Ar/F_2/Xe at a total pressure of 3.4 atm. The coupled framework enables simultaneous computation of discharge characteristics, species kinetics, photons, and laser power. The study focuses on the influence of F_2 concentration, aiming to predict its optimal value for maximizing laser output. These insights provide a theoretical foundation for guiding gas mixture optimization and enhancing the performance and stability of ArF excimer lasers in advanced applications.

II. MODEL DESCRIPTION

To simulate the discharge characteristics of the ArF excimer laser, a 1D fluid model coupled with circuit and laser modules is established and solved by using COMSOL Multiphysics,[18] which is schematized in Fig. 1. The configurations of the circuit, fluid, and laser modules, as well as the coupling mechanisms between them, are described in Sec. II A, Circuit module, Sec. II B, 1D fluid model, and Sec. II C, Laser gain module, respectively.

A. Circuit module

The pulsed discharge of an excimer laser is typically excited using magnetic pulse compression (MPC) technology.[19,20] The MPC network consists of two to four cascaded CLC loops, through which energy is progressively transferred between capacitive stages, compressing the voltage rise time to approximately 100 ns. In the developed circuit module, the last-stage MPC loop is incorporated and connected to the discharge electrodes, as illustrated in Fig. 1.

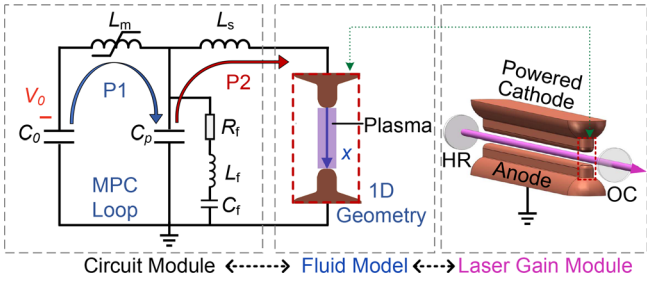


FIG. 1. Schematics of the mutually coupled circuit, fluid, and laser modules for the established 1D model.

The circuit module comprises three primary loops: the MPC loop, an RLC filter loop, and the discharge load loop. The MPC loop consists of an energy storage capacitor C_0, a magnetic switch L_m, and a peaking capacitor C_p. The energy storage capacitor C_0 is pre-charged to an initial voltage V_0 at $t = 0$. The magnetic switch L_m is set as a saturable inductor with a low inductance. The component values are specified as $C_0 = 10$ nF, $L_m = 250$ nH, $C_p = 10$ nF, and $V_0 = -18$ kV.

The discharge load loop includes C_p, a discharge stray inductor $L_s = 5$ nH, and the discharge gap. To suppress circuit oscillations after gap breakdown, an RLC filter branch is incorporated with the following parameters: $R_f = 70.71\ \Omega$, $L_f = 100$ nH, and $L_f = 10$ pF.

The operation of the circuit module involves two distinct phases. During Phase 1 (P1), energy is transferred from the capacitor C_0 to C_p. In Phase 2 (P2), once the electrode voltage reaches the breakdown threshold, C_p rapidly discharges its stored energy into the discharge gap.

In the developed model, the circuit module and the fluid module are bidirectionally coupled and iterated simultaneously. The circuit module, governed by ordinary differential equations (ODEs), computes the instantaneous voltage U_{ext} applied to the electrodes. This voltage serves as the Dirichlet boundary condition for solving Poisson's equation within the plasma domain,

$$\nabla \cdot (\varepsilon_r \varepsilon_0 \nabla \phi_{el}) = -\rho_c \quad \text{with} \quad \phi_{el} = U_{ext},$$

where ε_r and ε_0 denote the relative permittivity, ϕ_{el} is the electric potential, and ρ_c is the net space charge density.

Moreover, the electrode current I_{el} is computed at the electrode surface as the sum of conduction and displacement currents,

$$I_{el} = \int_{el} \left(\sum_j q_j \Gamma_j \cdot \hat{n} + \frac{\partial(\varepsilon_r \varepsilon_0 \mathbf{E})}{\partial t} \cdot \hat{n} \right) dA_{el},$$

where the first and second terms represent the conduction current and the displacement current, respectively. q_j is the charge of species j, Γ_j is the flux density of species j, $\hat{n}$ is the unit normal vector pointing into the electrode, $\mathbf{E}$ is the electric field, A_{el} is the area of the electrode.

B. 1D fluid model

This study employs a one-dimensional fluid model, based on the drift-diffusion approximation, to describe plasma behavior along the interelectrode axis ($d = 12$ mm), as illustrated in Fig. 1. The density n_j of each particle species j in Table II, except for UV photons, is governed by the continuity equation,

$$\frac{\partial n_j}{\partial t} + \nabla \cdot \Gamma_j = S_j, \tag{1}$$

where S_j is the source term and Γ_j is the particle flux. The control equation for the photon density will be introduced in the laser gain module. The particle flux Γ_j is given by

$$\Gamma_j = -D_j \nabla n_j + z_j \mu_j n_j \mathbf{E}, \tag{2}$$

where z_j is the charge of species j. D_j and μ_j are the diffusion and mobility coefficients, respectively, computed using BOLSIG+.[21]

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The electron energy density n_ϵ is described by the electron energy conservation equation,

$$\frac{\partial n_\epsilon}{\partial t} + \nabla \cdot \Gamma_\epsilon + e\mathbf{E} \cdot \Gamma_e = S_\epsilon, \tag{3}$$

where S_ϵ represents the energy gain from electron collisions, and the energy flux Γ_ϵ is expressed as

$$\Gamma_\epsilon = -D_\epsilon \nabla n_\epsilon - \mu_\epsilon n_\epsilon \mathbf{E} \tag{4}$$

with the electric field $\mathbf{E}$ determined by Poisson's equation,

$$\nabla \cdot \mathbf{E} = -\frac{(n_i - n_e)e}{\varepsilon_r \varepsilon_0}, \tag{5}$$

where ε_r and ε_0 denote the relative permittivity and vacuum permittivity, respectively.

The ArF excimer laser operates with a working gas mixture of Ne/Ar/F_2/Xe, with an initial ratio of 96.4/3.5/0.1/0.01 at a total pressure of 3.4 atm. The model includes 26 distinct particle species (see Table I) and accounts for 41 plasma reactions (detailed in Appendix). The heavy-particle reaction rate constants are primarily obtained from Ma *et al.*[13] The electron reaction cross sections are sourced from the LXCat database[22] and calculated using the Boltzmann solver BOLSIG+.[21] BOLSIG+ is only calculated once at the beginning of the computation, which comprehensively provides the precomputed data (electron transport parameters and electron-collision reaction rates) as functions of electron energy (eV). The detailed plasma chemical reactions are provided in the Appendix. The 193 nm photons are emitted through both spontaneous and stimulated emission processes from ArF (B), the upper laser level of the excimer molecule. For simplicity, ArF^* is used to denote the excited-state ArF (B). The ground-state ArF (X) is not considered in the model because its dissociation time is on the order of picoseconds and is therefore negligible. Therefore, the de-excitation of ArF (B) directly leads to its dissociation into Ar and F atoms, as shown in ArF quenching and radiation reaction groups in the Appendix.

Figure 2 illustrates the primary chemical reactions involved in the operation of an ArF excimer laser. The contribution of each reaction pathway is visually represented by arrow thickness, where increased thickness denotes greater pathway contribution. The ionic channel ($Ar^+ + F^-$) and the neutral channel ($Ar^* + F_2$) are the dominant pumping reaction channels for ArF^* formation. The ArF^* quenching reaction channels include electron-impact dissociation, two-body collision dissociation (with Ne, Ar, and F_2), and three-body collision reactions (with Ne and Ar). The laser reactions, comprising spontaneous excitation and stimulated emission, potentially compete with the ArF^* quenching reactions. The dissociation and attachment reactions of F_2 influence the densities of F^- and F_2. These reactions will be focused on in the simulation.

TABLE I. List of plasma species.

Type	Species
Neutral species	Ar, F_2, Ne, Xe, F
Ar excited species	Ar^*, Ar^{**}, Ar_2^*
Ne excited species	Ne^*, Ne^{**}, Ne_2^*
F_2 excited species	$F_2(A^1)$, $F_2(a^3)$
Xe excited species	Xe^*
Dimer species	ArF^*, Ar_2F^*
Positive charged species	Ar^+, F_2^+, Ne^+, Xe^+, Ar_2^+, Ne_2^+
Negative charged species	F^-, e
UV Photons	$h\nu_{193}$ (ArF), $h\nu_{95}$ (Ar^{**})

The pre-ionization process (sufficient initial electron density $n_{e,0}$) is a necessary condition for achieving uniform breakdown at a high pressure,[24,25] which promotes improved laser beam power and quality. The pre-ionization is typically achieved using corona rods positioned near the cathode,[26] which generate an initial electron density on the order of 10^{13} m^{-3}, as documented in Refs. 27 and 28. Since the pre-ionization occurs near the cathode, the initial distribution of pre-ionization electron density $n_{e,0}$ exhibits a certain attenuation from the cathode to the anode. In Fig. 3, the initial electron density $n_{e,0}$ at $t = 0$ is set according to the following equation, with the density gradually increasing from 10^{13} m^{-3} near the anode to 10^{14} m^{-3} near the cathode:

$$n_{e,0}(x) = 10^{14} \times \tanh(100x) + 10^{13}\ m^{-3}. \tag{6}$$

The discharge parameters implemented in this model are provided in Table II.

C. Laser gain module

As the fluid model cannot simulate photon density, a custom laser gain module using the PDE interface is developed within COMSOL. Based on the following equation,[29,30] this module calculates the temporal variation of the photon density dn_{ph}/dt and is coupled with the fluid model:

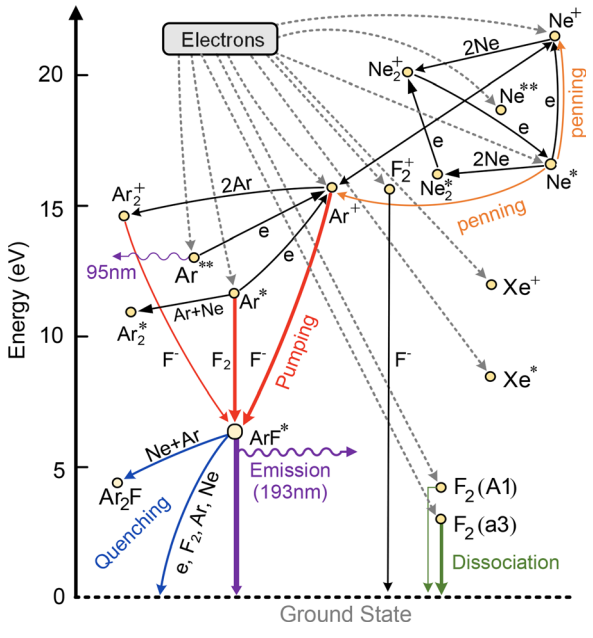


FIG. 2. Schematic diagram of ArF laser discharge reactions in a Ne/Ar/F_2/Xe gas mixture.[23] The contribution of each reaction pathway is visually represented by arrow thickness, where increased thickness denotes greater pathway contribution.

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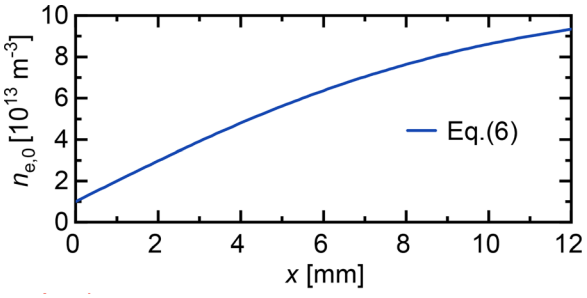


FIG. 3. Initial electron density distribution $n_{e,0}$ along the x axis. The anode is located at $x = 0$ mm, while the cathode is located at $x = 12$ mm.

$$\frac{dn_{ph}}{dt} = \frac{l_g}{l_c}(R_{STE} - R_\alpha) + K \cdot R_{SE} - L, \tag{7}$$

where l_g is the resonator length, l_c is the cavity length. R_{STE}, R_α, R_{SE} are the reaction rates of stimulated emission (STE), photon absorption, and spontaneous emission (SE), respectively. These photon-involved reactions are considered in the plasma fluid model, and their reaction rates are calculated based on the following equations:

$$R_{STE} = n_{ph} n_{ArF^*} \sigma_{38},$$
$$R_\alpha = n_{ph}(n_{Ar^{**}} \sigma_{R40} + n_{F^-} \sigma_{R41}),$$
$$R_{SE} = \frac{n_{ArF^*}}{\tau},$$

where n_{ArF^*}, $n_{Ar^{**}}$, and n_{F^-} are the densities of ArF^*, Ar^**, and F^-, respectively. σ_{38} is the cross section for the stimulated emission reaction, while σ_{40} and σ_{41} correspond to the two photon-absorbing reactions. τ is the lifetime of ArF^*. According to Eq. (7), the photon density n_{ph} in the laser module is determined by the reaction rates R_{STE}, R_α, and R_{SE} from the plasma module, which in turn are functions of n_{ph}, thereby coupling the two modules.

In Eq. (7), K is the probability of generating photons that move along the axis via the SE. Since the SE is an isotropic process, the probability K of generating photons that move along the cavity axis is very small, with a value of $K = 10^{-5}$. L is the optical cavity loss rate, which is given by

$$L = \frac{cn_{ph}}{2l_c}\left[\ln R_{OC} + \ln(1 - L_{op})\right], \tag{8}$$

where c is the speed of light. R_{OC} is the output coupler reflectivity. Assuming 100% reflectivity for the HR mirror, the loss at the HR mirror is neglected in the model. L_{op} is the single-pass optical loss factor, which reflects the proportion of losses due to non-output coupling, including absorption losses, scattering losses, and so on. The above constant parameters are set as $l_c = 105$ cm, $l_g = 90$ cm, $K = 10^{-5}$, $R_{OC} = 8\%$. In Eq. (8), the single-pass optical loss factor L_{op} is set as a variable dependent on the F_2 concentration. This is because an increase in F_2 concentration leads to a degradation of discharge uniformity, resulting in an increase in the laser's intra-cavity loss.[12] The loss factor L_{op} is set to be linearly related to the F_2 concentration and is given by

$$L_{op} = k_c \times C_{F_2}, \tag{9}$$

where the loss factor L_{op} is typically set to 0.08 for an F_2 concentration of 0.1%.[29,30] Thus, the slope k_c is set to 80.

Based on laser principles,[31,32] the output power of the laser is expressed as

$$P_L = n_{ph} ch\nu_{193} S \eta_{out}, \tag{10}$$

where h is Planck's constant, ν_{193} is the frequency of the 193 nm photon, S is the cross-sectional area of the laser beam, and η_{out} is the output coupling efficiency, defined as the ratio of the output intensity I_{out} to the average intra-cavity intensity $\overline{I_{in}}$ with

$$\eta_{out} = \frac{I_{out}}{\overline{I_{in}}}. \tag{11}$$

In previous studies,[33] the output coupling efficiency of continuous-wave lasers ($\eta_{out} = 0.5$) has occasionally been adopted for estimating output power in pulsed-mode lasers, which leads to prediction errors. To address this, this study presents a derivation of the output power expression for excimer lasers in pulsed mode that can be computed in real time using the established fluid–laser gain model. Figure 4 depicts the amplification of the forward (I_+) and backward (I_-) intensities within the laser cavity. Assuming the laser gain grows exponentially along the propagation path, the forward and backward intensities at any position $z \in [0, l_g]$ can be expressed as

$$I_+(z) = I_1 \exp(gz), \tag{12}$$

$$I_-(z) = I_3 \exp(gz). \tag{13}$$

Here, the gain coefficient g is determined in real time by the rates of stimulated emission and photon absorption reactions, given by

$$g(m^{-1}) = n_{ArF^*} \sigma_{R38} - (n_{Ar^{**}} \sigma_{R40} + n_{F^-} \sigma_{R41}). \tag{14}$$

The average intra-cavity intensity is then obtained as

TABLE II. List of discharge parameters.

Parameter	Variable symbol	Value
Applied voltage	V_0	− 18 kV
Gas pressure	p_0	3.4 atm
Gas temperature	T_g	300 K
Gap distance	d	12 mm
Discharge area	A	90 cm^2
Initial electron density	$n_{e,0}$	$n_{e,0}(x)$

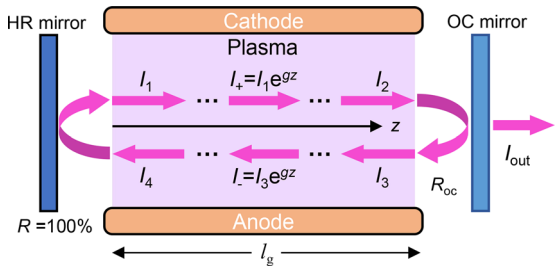


FIG. 4. Schematics of the amplification of the forward (I_+) and backward (I_-) intensities within the laser cavity.

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$$\overline{I_{\text{in}}} = \frac{1}{l_g}\int_0^{l_g}[I_+(z) + I_-(z)]\mathrm{d}z = I_1\frac{[1+\exp(gl_g)R_{\text{oc}}][\exp(gl_g)-1]}{gl_g}. \quad (15)$$

As shown in Fig. 4, the boundary conditions at the OC mirror are given by

$$I_2 = I_1\exp(gl_g), \quad (16)$$

$$I_{\text{out}} = I_2(1-R_{\text{oc}}) = I_1\exp(gl_g)(1-R_{\text{oc}}). \quad (17)$$

Finally, by combining Eqs. (15) and (17) to eliminate the intermediate variable I_1, the output coupling efficiency η_{out} is derived as follows; thus, the output power P_{L} is evaluated dynamically based on Eq. (10):

$$\eta_{\text{out}} = \frac{gl_g\ \exp(gl_g)(1-R_{\text{OC}})}{[1+\exp(gl_g)R_{\text{OC}}][\exp(gl_g)-1]}. \quad (18)$$

D. Model validation

To validate the 1D model, Fig. 5 compares the measured (solid lines) and simulated (dashed lines) voltage and current pulse waveforms. Both the experiment and simulation are performed with an identical Ne/Ar/F_2/Xe gas mixture at a pressure of 2 atm. Due to experimental constraints, the working pressure and circuit parameters in this validation differ from those described in Sec. II A, and the OC and HR mirrors are omitted. For model validation, the simulation parameters in this subsection are set to match the experimental conditions. In Sec. III, however, they are identical to those in the Circuit Module to reflect industrial conditions. The circuit parameters used for both the experiment and simulation are as follows: $C_0 = 10.4$ nF, $C_{\text{p}} = 8.1$ nF, $V_0 = -13$ kV, a discharge stray inductance of $L_{\text{d}} = 30$ nH, a gap distance of $d = 20$ mm, and a repetition rate of 5 Hz. The magnetic switch L_{m} transitions from the off-state to the on-state when the current exceeds 200 A, with its inductance changing from 5 to 1 μ H. The close agreement between the simulated and experimental waveforms confirms the validity of the proposed model.

III. RESULTS AND DISCUSSION

A. Fundamental dynamic characteristics

In the following, the results from a base case gas mixture of Ne/Ar/F_2/Xe at an F_2 concentration of 0.1% are presented. Figure 6(a) shows the waveforms of the gap voltage U_{gap}, discharge current I_{dis}, and laser power P_{L} at an F_2 concentration of 0.1%. The simulated laser power waveform exhibits a full width at half maximum (FWHM) of 15 ns and reaches a peak value of 5 MW. The results are compared with those from the two-dimensional (2D) model (dashed curves) developed by Ma *et al.*,[13] under identical conditions including the mixture ratio, gas pressure p_0, gap distance d, and initial charging voltage V_0. The value of breakdown voltage and the trend of pulse oscillation show good agreement with the reference data, demonstrating the reliability of the proposed 1D model. Additionally, a time delay is observed between the laser pulse and the discharge current pulse. This observation is consistent with the simulation results reported by Akashi *et al.*[30] The underlying mechanism is that during the current peak, a large number of ArF* molecules are produced, but the photon density remains low. As a result, the reaction rate of the STE process ($\text{ArF}^* + h\nu \rightarrow 2h\nu + \text{Ar} + \text{F}$) is relatively low. After a short delay, the SE process ($\text{ArF}^* \rightarrow h\nu + \text{Ar} + \text{F}$) generates sufficient photon, which in turn enhances the STE process and leads to a laser pulse.

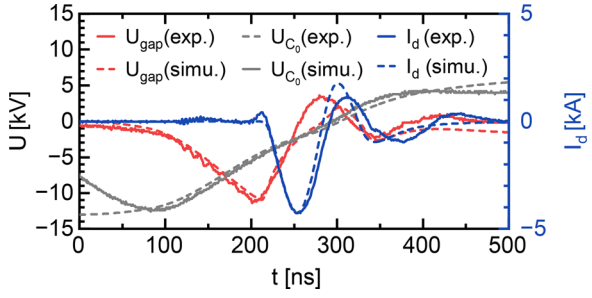


FIG. 5. Comparison of voltage and current pulse waveforms obtained from experimental measurements (solid curves) and simulations (dashed curves).

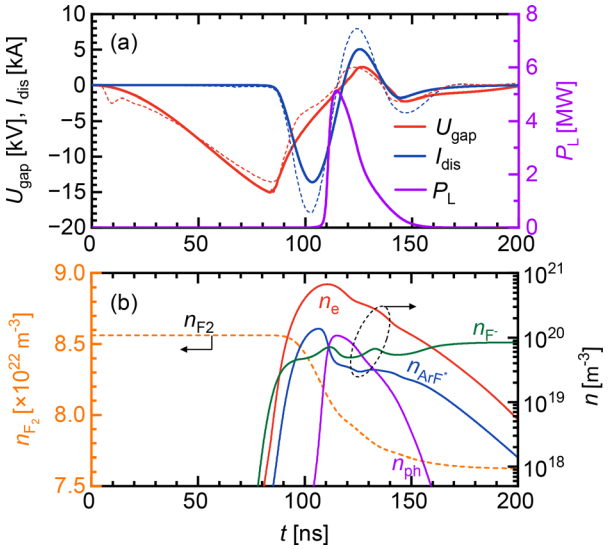


FIG. 6. (a) Waveforms of the gap voltage U_{gap}, discharge current I_{dis}, and laser power P_{L} at an F_2 concentration of 0.1%. The solid curves represent the simulation results from our model, while the dashed curves are extracted from Ma *et al.*[13] (b) Temporal evolutions of the number densities of dominant species.

Figure 6(b) shows the temporal evolutions of the number densities of dominant species at an F_2 concentration of 0.1%. The density of F_2 is shown on the left axis, while the densities of the other particles (e, ArF*, F^-, and photons at 193 nm) are shown on the right axis. The consumption of F_2 is evident among the four initial neutral gases, as evidenced by studies on gas lifetime limitations and performance degradation in ArF and KrF systems.[34–36] When the gap breaks down, the density of F_2 rapidly decreases with F^- ions increasing. Notably, the ArF* density increases with increasing electron density, and their peaks align closely in time. At $t = 108$ ns, the density of ArF* reaches its peak. Subsequently, the density of ArF* drops rapidly via the STE process with the production of a large number of 193 nm photons along the laser axis. The photon density rises to a maximum of 10^{20} m^{-3} within about 10 ns before sharply declining to 10^{18} m^{-3} by

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approximately 45 ns. At the end of the discharge, the densities of F_2 and F^- tend to stabilize, while the densities of electrons, ArF^*, and 193 nm photons decrease to relatively low values.

Figure 7 shows temporal evolutions of the reaction rates for ArF^* formation and ArF^* loss reactions. The area formed by the reaction rate curve and the t-axis represents the ArF^* density via formation and quenching channels. The density of ArF^* decreases to a very low value at the end of the discharge, as shown in Fig. 6(b), so the total areas of regions in Figs. 7(a) and 7(b) are almost equal. It can be clearly seen that the neutral channel $Ar^* + F_2 \rightarrow ArF^* + F^-$ is the primary pathway for ArF^* formation, followed by the ionic channel $Ar^+ + F^- \rightarrow ArF^*$, while the ionic channel $Ar_2^+ + F^-$ has a relatively small contribution to ArF^* formation. The density of ArF^* quenched through each channel occupies a considerable proportion. In the initial breakdown phase ($t = 90 \sim 110$ ns), the SE process generates only a limited number of axially propagating 193 nm photons. When the photon density surpasses the critical threshold (at $t = 110$ ns), the STE process dominates, leading to rapid ArF^* depletion and accelerated photon emission with laser output. Although the SE process produces a large number of 193 nm photons, only a small fraction of photons propagate axially [according to Eq. (7)] about ratio $K = 10^{-5}$.

As the upper-level laser species, the contributions of different reactions to the production and consumption of ArF^* have been analyzed. The density of ArF^* plays an important role in the entire kinetic process. In Fig. 8, the changes in ArF^* density and photon density $\Delta n_{ArF^*,k}$, $\Delta n_{ph,k}$ from a specific reaction channel k are quantified as the width of the pipelines, calculated using the following equations:

$$\Delta n_{ArF^*,k} = N_A \int_0^{t_{tot}} R_k \mathrm{d}t, \tag{19}$$

$$\Delta n_{ph,k} = N_A \int_0^{t_{tot}} R_k \mathrm{d}t, \tag{20}$$

where N_A is the Avogadro constant and t_{tot} is the total discharge duration of 200 ns. The reaction rate R_k (mol · m^{-3}· s^{-1}) in the fluid model corresponds to R29–R31 for ArF^* formation, to R32–R39 for ArF^* loss—among which R37 and R38 represent the SE and STE processes,

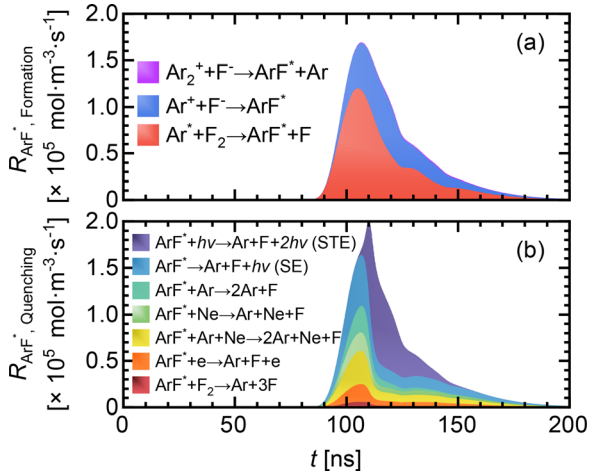


FIG. 7. Temporal evolutions of the reaction rates for (a) ArF^* formation and (b) ArF^* loss reactions.

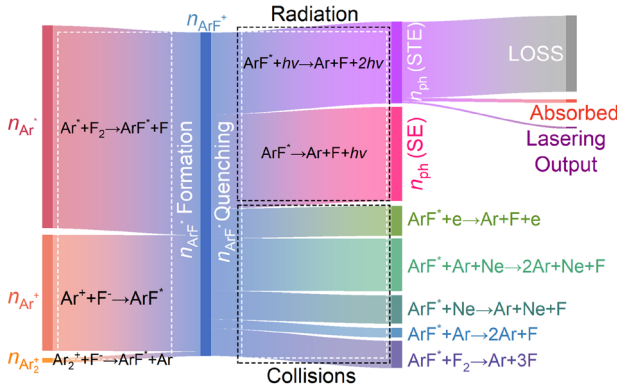


FIG. 8. Reaction pathways governing the population dynamics of ArF^* molecules and 193 nm photons. SE: Spontaneous emission. STE: Stimulated emission. The height of the node (the width of the pipe) represents the density of a certain particle changing through this reaction channel.

respectively—and to R40–R41 for photon absorption. It can be observed that the formation of ArF^* primarily relies on two key pathways: the neutral channel ($Ar^* + F_2$) and the ionic channel ($Ar^+ + F^-$). Approximately half of ArF^* is quenched via radiative processes (SE and STE), with the remainder attributed to collisional losses involving electrons and heavy particles. The SE process, which occurs in all directions, contributes very little to the output laser in the axial direction. The 193 nm photons produced via stimulated emission experience significant cavity losses [characterized by the optical loss rate L in Eq. (7)], with merely 2.22% of the total photon population contributing to the effective laser output (P_L).

B. Effects of F_2 concentration

The above simulation results show that the electronegative gas F_2 plays an important role in the generation and consumption of ArF^* and 193 nm photons. In Sec. IV, the influence of F_2 concentration on both discharge characteristics and laser performance is systematically investigated. The F_2 concentration is varied from 0.05% to 0.5% with an increment of 0.0167%, covering a total of 28 cases. For each case, the electron transport parameters and electron collision reaction rates are recalculated using BOLSIG+ and subsequently imported into the 1D fluid model.

Figure 9 shows the ionization coefficient α (in m^{-1}) and attachment coefficient η (in m^{-1}) as a function of the reduced electric field

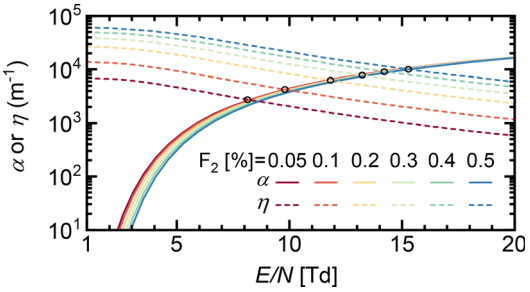


FIG. 9. Ionization coefficient α (in m^{-1}) and attachment coefficient η (in m^{-1}) as a function of the reduced electric field E/N (in Td) at different F_2 concentrations.

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E/N (in Td) at different F_2 concentrations. The electron ionization coefficient and attachment coefficient are calculated using the Boltzmann solver BOLSIG+[21] based on electron collision reaction cross sections (R1–R17 in the Appendix). Fors all F_2 concentrations, the reduced electric field E/N predominantly varies within the range of $1 \sim 20$ throughout the pulse discharge cycle (see Fig. 10), which is adopted as the range of the horizontal axis. As the F_2 concentration increases, the coefficient α remains essentially unchanged, while η continues to strengthen. The rightward shift of the intersection point indicates that the electric field strength required for effective electron multiplication is increasing. This suggests that the breakdown voltage and electron temperature may increase as the F_2 concentration rises. These results are critical for understanding the breakdown and discharge characteristics in fluorine-based excimer laser discharge systems.

Figure 10(a) shows the temporal evolution of electron density n_e at F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%. The maximum electron density is shown in the order of magnitude 10^{20} m^{-3} under these F_2 concentrations. As the F_2 concentration increases, the electron density continues to decrease. This is because the effective ionization coefficient $(\alpha - \eta)$ keeps decreasing as the F_2 concentration increases. To achieve gap breakdown under conditions of low effective ionization coefficients $(\alpha - \eta)$, an increased breakdown field strength (peak reduced electric field E/N) is required, as shown in Fig. 10(b). Figure 10(c) shows the temporal evolution of the mean electron temperature T_e for different F_2 concentrations. T_e remains relatively low

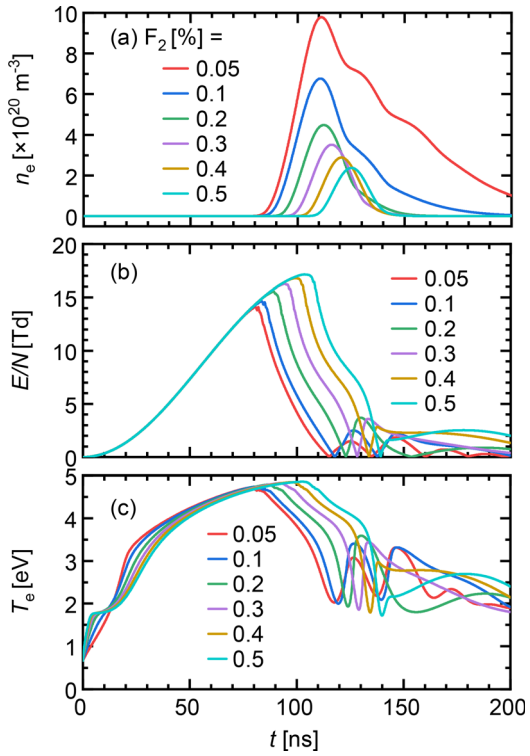


FIG. 10. Temporal evolution of (a) electron density n_e, (b) reduced electric field E/N, and (c) mean electron temperature T_e at F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%.

throughout the discharge, varying within a range of 1–5 eV. A higher F_2 concentration results in a greater initial T_e at $t = 10$ ns, which is attributable to the enhanced absorption of low-energy electrons by F_2.[16] Furthermore, the peak T_e value exhibits a moderate increase with F_2 concentration, consistent with the higher breakdown electric field required for ignition.

To further understand the electronegative mechanism of F_2 in ArF mixed gas discharge, the loss characteristics of F_2 molecules are studied based on key reaction channels. As the F_2 concentration increases, the pressure drop caused by the consumption of F_2 increases from 0.34 to 0.5 mbar, as shown in Fig. 11. The F_2 loss density ($n_{F_2,k}$, m^{-3}) is determined through time integration of the reaction rates, following a computational approach similar to Eq. (19).

Figure 11 illustrates the F_2 molecular loss density (left axis, $\times 10^{22}$ m^{-3}) via dominant consumption pathways as functions of F_2 concentration. The electron-impact dissociation process, $e + F_2(a^3) \rightarrow e + 2F$, represents the primary F_2 consumption mechanism, with the ArF^* formation channel ($Ar^* + F_2 \rightarrow ArF^* + F$) serving as the secondary pathway. The reaction rates involving F_2 dissociation (via A^1 state), F_2 ionization, and recombination reactions are negligibly small and have been omitted for clarity. The loss of F_2 via both electron-impact dissociation and ArF^* quenching channels increases with increasing F_2 concentration. Though the attachment coefficient η increases with rising F_2 concentration (see Fig. 9), F_2 loss density via the attachment reaction ($e + F_2 \rightarrow F^- + F$) remains nearly constant. This phenomenon may be attributed to the enhanced reduced electric field E/N at higher F_2 concentrations, which promotes the dissociation of F_2 into the a^3 state while suppressing low-energy electron attachment.

Figure 12 compares the reaction rates of electron attachment-dissociation ($e + F_2 \rightarrow F + F^-$) and direct electron-impact dissociation [$e + F_2(a^3) \rightarrow e + 2F$] with F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%. As the F_2 concentration increases, both reactions demonstrate enhanced peak rates. However, the attachment-dissociation channel exhibits progressively narrower temporal profiles at higher F_2 concentrations, which results in a nearly constant time-integrated F_2 loss density, consistent with the observations in Fig. 11.

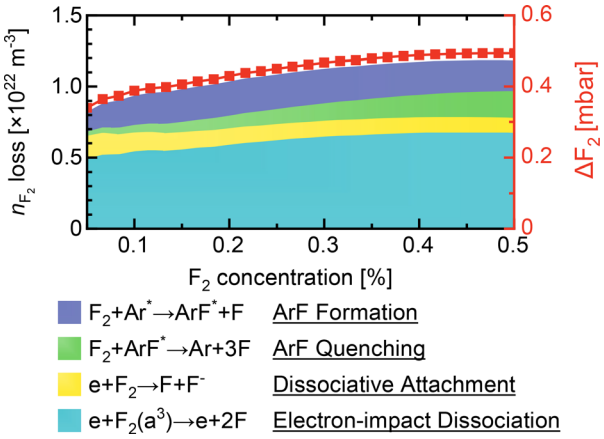


FIG. 11. F_2 loss density (left axis, $\times 10^{22}$ m^{-3}) via dominant consumption pathways and total pressure depletion (right axis, mbar) as functions of F_2 concentration.

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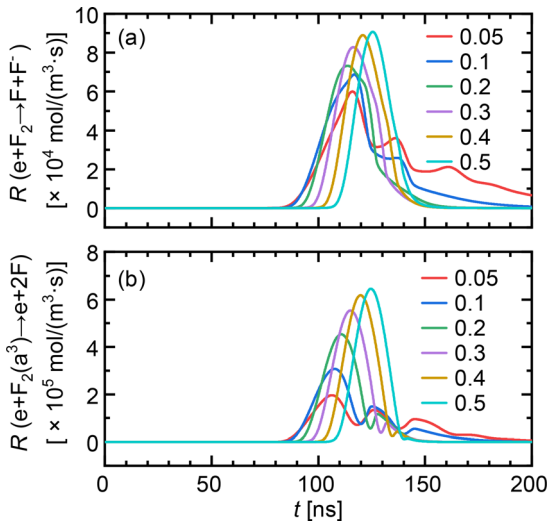


FIG. 12. Temporal evolution of reaction rates for (a) electron attachment-dissociation ($e + F_2 \rightarrow F + F^-$) and (b) direct electron-impact dissociation [$e + F_2(a^3) \rightarrow e + 2F$] with F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%.

The dynamics of ArF* are of particular focus, as they directly determine the intensity of the stimulated photon emission. Figure 13 displays the temporal evolution of the total reaction rates for the major ArF formation and quenching channels at F_2 concentrations of (a) 0.05% (b) 0.2%, and (c) 0.5%. The overall rates of ArF* generation and quenching first increase and then decrease as the F_2 concentration rises. The STE process begins as the SE rate starts to decline from its peak. At the moderate F_2 concentration of 0.2%, the STE rate exhibits a sharp, distinct peak, which is highly favorable for generating a laser pulse with a short FWHM. In contrast, at both low (0.05%) and high (0.5%) F_2 concentrations, the SE and STE reactions are suppressed and delayed.

Figures 14(a) and 14(b) illustrate the density of ArF* produced through the generation channels and the density of ArF* lost through the loss channels with increasing F_2 concentrations, respectively. Since the density of ArF* drops to 0 at the end of the discharge, the total production density of the ArF* generation channel is almost equal to the total loss of the quenching channels. It can be seen that the ion channel $Ar_2^+ + F^- \rightarrow ArF^* + Ar$ produces very little ArF*. The neutral channel $Ar^* + F_2 \rightarrow ArF^* + F$ is the dominant pathway for ArF* formation compared to the ion channel $Ar^+ + F^- \rightarrow ArF^*$. As the F_2 concentration increases, the ArF* density produced via these reaction channels initially increases and then decreases. As for the loss of ArF*, the F_2 quenching channel ($ArF^* + F_2$) gradually becomes dominant with increasing F_2 concentration. Since the reaction rate of spontaneous emission (SE) is determined by ArF* density, the SE process is shown to be inhibited slightly at high F_2 concentrations. According to Eq. (7), it can be deduced that within a certain period of time t, the number density of 193 nm photons n_{ph} increases exponentially with the density of ArF*, i.e., $n_{ph} \sim e^{n_{ArF^*}} t$. Therefore, the STE process is rapidly enhanced due to the increase in ArF* density before the concentration of F_2 reaches about 0.2%. However, when the F_2

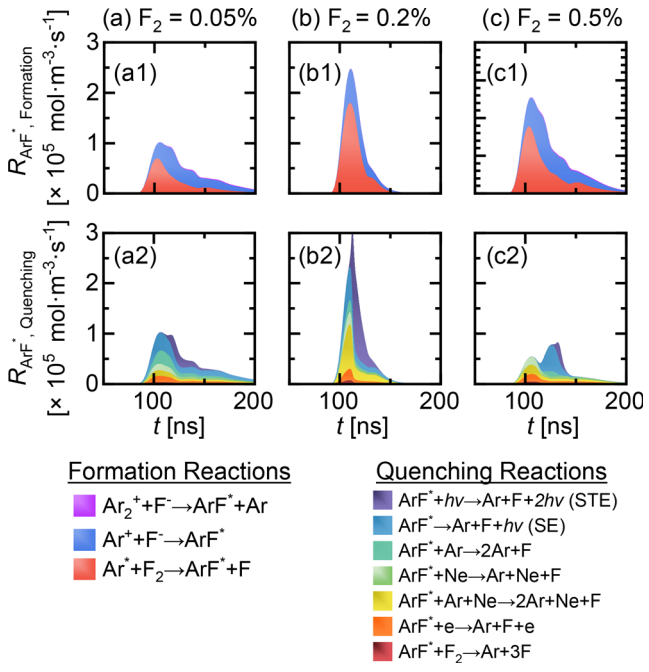


FIG. 13. Temporal evolution of reaction rates of the ArF formation and quenching reactions at F_2 concentrations of (a) 0.05%, (b) 0.2%, and (c) 0.5%.

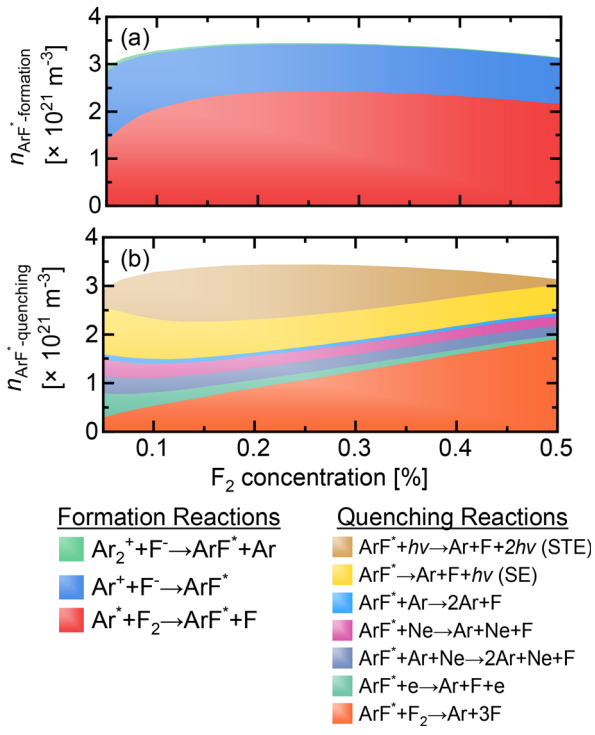


FIG. 14. (a) Formation and (b) quenching of ArF* number density (m^{-3}) via various pathways with F_2 concentration varying from 0.05% to 0.5%.

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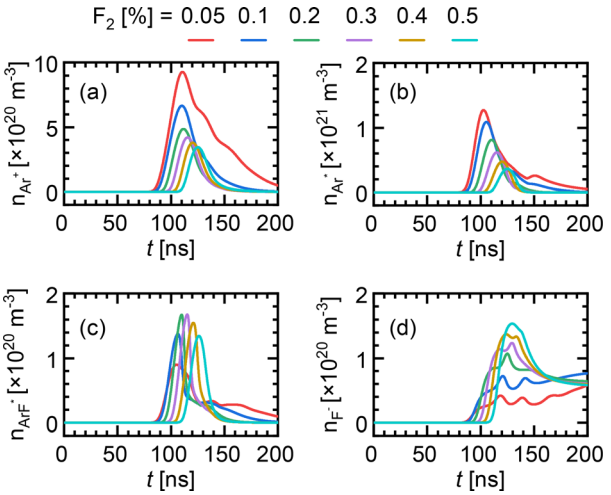


FIG. 15. Temporal evolution of number densities for (a) Ar^+, (b) Ar^*, (c) ArF^*, and (d) F^- with F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%.

concentration continues to rise, the STE process is rapidly suppressed due to the decline in ArF^* density.

Figure 15 presents the temporal evolution of number densities for Ar^+, Ar^*, ArF^*, and F^- with F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%. The results demonstrate that increasing F_2 concentration leads to a gradual reduction in Ar^+ and Ar^* densities. This is because the enhanced electron attachment effect suppresses ionization and excitation processes. The reduction in these two species would normally limit ArF^* formation but is counterbalanced by increased availability of F^- and F_2, which promote ArF^* formation. These competing effects result in a non-monotonic variation of ArF^* density, which initially increases and then decreases with increasing F_2 concentration. At F_2 concentrations of 0.2% and 0.3%, the density of ArF^* reaches relatively high peaks.

To further investigate the external characteristics of the laser under different F_2 concentrations, the performance of the discharge pulse and laser power is studied next. Figure 16(a) shows the waveforms of the gap voltage U_{gap} for F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, and 0.5%. As the F_2 concentration increases, the breakdown voltage rises significantly. The primary reason lies in the reduction of the effective ionization coefficient $\alpha - \eta$, which necessitates a higher applied voltage and electric field strength to achieve breakdown.

Figure 16(b) shows the waveforms of discharge current I_{dis} for different F_2 concentrations. As the F_2 concentration increases, the peak current decreases. The timing of the current peak is delayed due to the delayed breakdown caused by the higher breakdown voltage at elevated F_2 levels.

As the F_2 concentration increases, the damping characteristics of the discharge pulse gradually change from underdamped to overdamped with a decrease in the number of oscillations. This indicates that the equivalent discharge impedance R_d increases with increasing F_2 concentration. The equivalent discharge impedance R_d can be computed as[30,37]

$$R_d = \frac{d}{eA\mu_e n_e}, \tag{21}$$

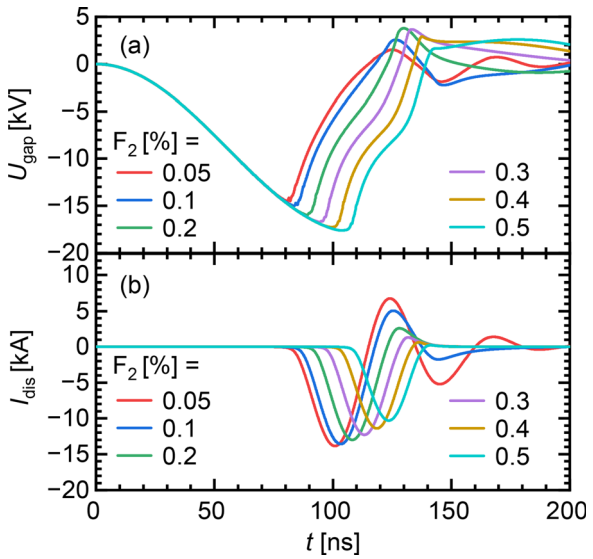


FIG. 16. Waveforms of (a) gap voltage U_{gap} and (b) discharge current I_{dis} for F_2 concentrations of 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, and 0.5%.

where e, μ_e, and n_e are the elementary charge, electron mobility, and electron density, respectively. d is the gap distance between the electrodes and A is the discharge area. Thus, the increase in discharge impedance is mainly attributed to the increase in electron density [see Fig. 10(a)] when the F_2 concentration rises. The decreasing current peak can also be attributed to the gradually increasing discharge impedance R_d.

Figure 17(a) shows the waveforms of the average discharge power density P_{dis} for varying F_2 concentrations. The average discharge power

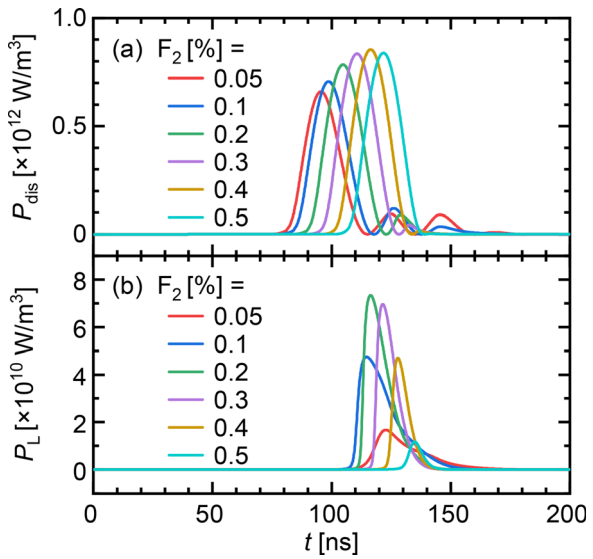


FIG. 17. Waveforms of (a) average discharge power density P_{dis} and (b) average laser power density P_L for different F_2 concentrations.

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density P_{dis} is calculated by dividing the spatially averaged power density along the x axis by the discharge volume V. As F_2 concentration increases, P_{dis} rises significantly. This increase is primarily driven by the higher breakdown voltage U_{gap} at elevated F_2 levels, while the discharge current I_{dis} remains relatively stable. The enhanced discharge power density reflects the greater energy deposition into the gas mixture, which is critical for sustaining the discharge and improving the laser power. Figure 17(b) shows the waveforms of the average discharge power density P_L at 193 nm for different F_2 concentrations. As the F_2 concentration increases, the peak value of P_L first increases and then decreases. The laser pulse is generally delayed with increasing F_2 concentration, consistent with the observed breakdown delay. However, an anomalous earlier laser pulse is observed at 0.05% F_2 concentration, even while the ArF density peak is delayed. This phenomenon is attributed to a stimulated emission depletion feedback mechanism. The stimulated emission rate, given by $R_{ste} = n_{ph} \times n_{ArF} \times \sigma$, declines with reduced ArF density under low F_2 concentration conditions. This suppression of R_{ste} diminishes photon generation, resulting in a positive feedback loop that further inhibits stimulated emission. Consequently, below a critical F_2 concentration threshold, the inadequate ArF density results in a significant delay in laser output.

The variations in the laser pulse FWHM (in ns) and conversion efficiency (CE) are further determined as a function of F_2 concentration, ranging from 0.033% to 0.5% in steps of 0.0167%, as illustrated in Fig. 18. The CE is defined as the ratio of the total laser energy (E_L) to the input energy ($E_{in} = 0.5C_0U_0^2 = 1.62$ J), where E_L is obtained by integrating the laser power over time. As the F_2 concentration increases, the FWHM of the laser pulse consistently decreases; the CE initially rises, reaching a maximum of approximately 5.8% at F_2 concentration = 0.15%, and then declines at higher F_2 concentrations. When the F_2 concentration is reduced to about 0.1% and below, the CE value drops rapidly. When the F_2 concentration reaches 0.033%, the CE drops to merely 0.2%, at which point lasing is nearly extinguished. The predicted optimal F_2 concentration agrees well with the experimental value of 0.22% reported by Lou *et al.*[10] for KrF excimer laser operation. The inflection point observed in the CE curve can be explained as follows. The ascending segment of the CE curve is attributed to an insufficient F_2 concentration, which limits the production of ArF^*. The descending segment results from a decline in the densities of Ar^* and Ar^+ [see Figs. 15(a) and 15(b)], which are the key reactants for ArF^* formation.

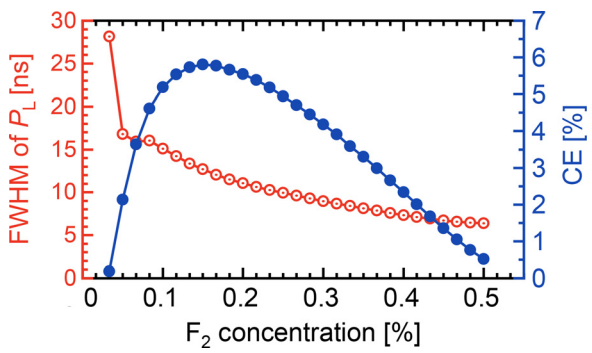


FIG. 18. Laser pulse FWHM and conversion efficiency (CE) as a function of F_2 concentration.

Additionally, increased photon losses due to discharge non-uniformity also contribute to the decrease in CE in this segment.

IV. CONCLUDING REMARKS

A one-dimensional fluid model, coupled with an external circuit and laser gain modules, is employed to study the kinetic characteristics of an ArF excimer laser across varying F_2 concentrations. It is found that as the F_2 concentration increases, the electron attachment coefficient α rises significantly, leading to a reduction in electron density n_e. This decline in the effective ionization coefficient $\alpha - \eta$ necessitates a higher reduced electric field E/N to achieve and sustain gas breakdown, thereby increasing the breakdown voltage. A higher reduced electric field implies that the discharge may tend to develop in a non-uniform and unstable direction. Moreover, an increase in F_2 concentration elevates the densities of reactants F_2 and F^- but reduces the densities of Ar^* and Ar^+, which are all crucial for ArF formation. As a result of this competition, the ArF density is maximized at an intermediate F_2 concentration, leading to an optimal laser power. The maximized power conversion efficiency is predicted by the model to be 5.8% at an F_2 concentration of 0.15%. For future work, the model will be extended to two dimensions to further explore the effects of electrode morphology and pre-ionization distribution on discharge uniformity.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jiayi Chen: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Liyang Zhang:** Formal analysis (equal); Investigation (equal); Validation (equal); Visualization (equal). **Zhihang Zhao:** Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal). **Yangyang Fu:** Conceptualization (lead); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Project administration (equal); Resources (lead); Supervision (lead); Visualization (equal); Writing – original draft (equal); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: REACTIONS INVOLVED IN THE MODEL

Reactions involved in the model. T_e is the electron temperature (eV) (Table III).

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TABLE III. Reactions involved in the model. T_e is the electron temperature (eV).

No.	Reaction process	Rate coefficient/cross section	Ref.
A. Electron collisions			
R1	$e + Ne \rightarrow Ne^* + e$	$f(\sigma)$	22
R2	$e + Ne \rightarrow Ne^{**} + e$	$f(\sigma)$	22
R3	$e + Ne \rightarrow Ne^+ + 2e$	$f(\sigma)$	22
R4	$e + Ne^* \rightarrow Ne^+ + 2e$	$f(\sigma)$	22
R5	$e + Ar \rightarrow Ar^* + e$	$f(\sigma)$	22
R6	$e + Ar \rightarrow Ar^{**} + e$	$f(\sigma)$	22
R7	$e + Ar \rightarrow Ar^+ + 2e$	$f(\sigma)$	22
R8	$e + Ar^* \rightarrow Ar^+ + 2e$	$f(\sigma)$	22
R9	$e + Ar^{**} \rightarrow Ar^+ + 2e$	$1.56 \times 10^{-13} T_e^{0.71} \exp(-2.63/T_e)\ m^3s^{-1}$	22
R10	$e + F_2 \rightarrow F^- + F$	$f(\sigma)$	22
R11	$e + F_2 \rightarrow F_2(a^3) + e$	$f(\sigma)$	22
R12	$e + F_2 \rightarrow F_2(A^1) + e$	$f(\sigma)$	22
R13	$e + F_2(a^3) \rightarrow 2F + e$	$2.08 \times 10^{-8} T_e^{-0.51} \exp(-4.69/T_e)\ m^3s^{-1}$	38
R14	$e + F_2(A^1) \rightarrow 2F + e$	$5.61 \times 10^{-9} T_e^{-0.23} \exp(-4.94/T_e)\ m^3s^{-1}$	38
R15	$e + F_2 \rightarrow F_2^+ + 2e$	$f(\sigma)$	22
R16	$e + Xe \rightarrow Xe^* + e$	$f(\sigma)$	22
R17	$e + Xe \rightarrow Xe^+ + 2e$	$f(\sigma)$	22
R18	$e + Ne_2^* \rightarrow Ne_2^+ + 2e$	$9.75 \times 10^{-15} T_e^{0.71} \exp(-3.4/T_e)\ m^3s^{-1}$	39
R19	$e + Ne_2^+ \rightarrow Ne + Ne^*$	$3.7 \times 10^{-14}\ T_e^{-0.5}\ m^3s^{-1}$	39
B. Penning ionization			
R20	$2Ne^* \rightarrow Ne^+ + Ne + e$	$5 \times 10^{-16}\ m^3s^{-1}$	13
R21	$Ne^* + Ar \rightarrow Ar^+ + Ne + e$	$6.5 \times 10^{-17}\ m^3s^{-1}$	40
B. Recombination			
R22	$F_2^+ + F^- \rightarrow 3F$	$4 \times 10^{-14}\ m^3s^{-1}$	13
C. Three-body heavy particle collisions			
R23	$Ar^* + Ar + Ne \rightarrow Ar_2^* + Ne$	$2.5 \times 10^{-43}\ m^6s^{-1}$	13
R24	$Ne^* + 2Ne \rightarrow Ne_2^* + Ne$	$4 \times 10^{-40}\ m^6s^{-1}$	39
D. Charge transfer			
R25	$Ne^+ + Ar \rightarrow Ar^+ + Ne$	$1 \times 10^{-17}\ m^3s^{-1}$	13
R26	$Ar^+ + Ar + Ne \rightarrow Ne^+ + 2Ar$	$1.6 \times 10^{-44}\ m^6s^{-1}$	13
R27	$Ar^+ + 2Ar \rightarrow Ar_2^+ + Ar$	$2.5 \times 10^{-43}\ m^6s^{-1}$	41
R28	$Ne^+ + 2Ne \rightarrow Ne_2^+ + Ne$	$4.4 \times 10^{-44}\ m^6s^{-1}$	39
D. ArF^* formation			
R29	$Ar^* + F_2 \rightarrow ArF^* + F$	$4 \times 10^{-16}\ m^3s^{-1}$	30
R30	$Ar^+ + F^- \rightarrow ArF^*$	$9.1 \times 10^{-13}\ m^3s^{-1}$	30
R31	$Ar_2^+ + F^- \rightarrow ArF^* + Ar$	$9.26 \times 10^{-13}\ m^3s^{-1}$	30
E.ArF^* quenching			
R32	$ArF^* + e \rightarrow Ar + F + e$	$2 \times 10^{-13}\ m^3s^{-1}$	41
R33	$ArF^* + F_2 \rightarrow Ar + 3F$	$1.9 \times 10^{-15}\ m^3s^{-1}$	41
R34	$ArF^* + Ar \rightarrow 2Ar + F$	$9 \times 10^{-18}\ m^3s^{-1}$	41
R35	$ArF^* + Ne \rightarrow Ar + Ne + F$	$1.6 \times 10^{-18}\ m^3s^{-1}$	13
R36	$ArF^* + Ar + Ne \rightarrow Ar_2F^* + Ne$	$3.5 \times 10^{-45}\ m^6s^{-1}$	13
F. UV radiation			
R37	$ArF^* \rightarrow Ar + F + h\nu_{193}$	$2.381 \times 10^8\ s^{-1}$ (spontaneous emission)	30
R38	$ArF^* + h\nu_{193} \rightarrow Ar + F + 2h\nu_{193}$	$4 \times 10^{-20}\ m^2$ (stimulated emission)	30
R39	$Ar^{**} \rightarrow Ar + h\nu_{95}$	$1 \times 10^9\ s^{-1}$	30
G. UV absorption (photoionization)			
R40	$Ar^{**} + h\nu_{193} \rightarrow Ar^+ + e$	$1 \times 10^{-22}\ m^2$	30
R41	$F^- + h\nu_{193} \rightarrow F + e$	$1 \times 10^{-21}\ m^2$	30

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